

OSD-767: Light Condition × Spaceflight Interaction Analysis

1. Introduction

The NASA OSD-767 experiment examined transcriptomic responses of tomato (*Solanum lycopersicum*) grown aboard the International Space Station under two light conditions (Red vs Blue LED lighting). Previous analysis using an additive DESeq2 model ($\sim \text{light} + \text{condition}$) identified 2,132 significant DEGs in leaf and 2,582 in root for the Flight vs Ground comparison. This report examines whether **light quality (Red vs Blue) modulates the spaceflight response** — i.e., whether the effect of microgravity on gene expression depends on the light environment.

This is biologically motivated: Red light is primarily sensed by phytochromes, while Blue light is sensed by cryptochromes and phototropins. These photoreceptor pathways intersect with stress signaling, circadian regulation, and hormone transduction — all pathways known to be altered during spaceflight. An interaction between light quality and spaceflight would suggest that light environment design could mitigate or exacerbate spaceflight stress responses.

Study Design

Tissue	Ground Red	Ground Blue	Flight Red	Flight Blue	Total
Leaf	6	6	5	4	21
Root	3	3	5	4	15
Combined	9	9	10	8	36

Key limitation: Root has only 3 ground replicates per light condition, reducing power for detecting interaction effects.

2. Methods

2.1 DESeq2 Interaction Models

Three DESeq2 models were fitted:

1. **Leaf interaction model** (21 samples): $\sim \text{light} + \text{condition} + \text{light}:\text{condition}$
2. **Root interaction model** (15 samples): $\sim \text{light} + \text{condition} + \text{light}:\text{condition}$
 - **Additive fallback** for root light main effect: $\sim \text{light} + \text{condition}$ (better powered)
3. **Combined whole-organ model** (36 samples): $\sim \text{tissue} + \text{light} + \text{condition} + \text{light}:\text{condition}$

Factor levels: `light` = {Red, Blue}, `condition` = {Ground, Flight}, `tissue` = {Leaf, Root}.

The interaction term `light:condition` tests whether the Flight-vs-Ground effect differs between Blue and Red light. A positive interaction LFC means the spaceflight response is stronger under Blue light; a negative interaction means it is stronger under Red light.

Significance thresholds:

- Main effects: $\text{padj} < 0.05$, $|\text{shrunk log2FC}| \geq 1$
- Interaction effects: $\text{padj} < 0.1$, $|\text{shrunk log2FC}| \geq 0.5$ (relaxed due to lower power for interactions)

LFC shrinkage was applied using `ashr` for all comparisons.

2.2 Gene ID Mapping

Solyc gene IDs (ITAG3.0/SL3.0 assembly) were mapped to Entrez Gene IDs via biomaRt (Ensembl Plants mart, dataset `slgca000188115v5cm_eg_gene`). Mapping rate: 59.7-75.0% across gene sets.

2.3 GO and KEGG Enrichment

- **GO enrichment:** `clusterProfiler::enricher()` with custom TERM2GENE/TERM2NAME from biomaRt GO annotations (no tomato OrgDb available). Background universe: 11,442 Entrez IDs. Analyzed BP, MF, CC separately.

- **KEGG enrichment:** `clusterProfiler::enrichKEGG(organism="sly")`. Background: all tomato genes in KEGG.
- Cutoffs: `pvalueCutoff=0.1`, `qvalueCutoff=0.2` (relaxed for interaction sets); `pvalueCutoff=0.3` for sets with <20 Entrez IDs.

2.4 Systems Biology Pathway Visualization

- **pathview** (R) was unavailable due to Rgraphviz compilation failure (graphviz API v2 vs v2.42 incompatibility).
- **SBGNview** (R) was unavailable for the same reason.
- **Alternative approach:** KEGG pathway diagrams were generated using:
 - KEGG REST API for reference images and KGML spatial layouts
 - Python bioservices + matplotlib for expression overlays
 - SBGN-style Process Description diagrams from KGML coordinates
 - Pathway-level LFC heatmaps and gene-level bar plots

2.5 Tissue Comparison

Leaf vs root comparisons included:

- Venn diagrams of overlapping gene sets
- Pearson/Spearman correlation of LFCs across tissues
- Shared vs tissue-specific KEGG/GO pathway analysis
- Directional consistency analysis

3. Results

3.1 Light Main Effect (Blue vs Red)

The light main effect captures genes differentially expressed between Blue and Red light, averaged across Ground and Flight conditions.

Model	Effect	Significant (padj<0.05, LFC >=1)	Up (Blue>Red)	Down (Blue<Red)
-----	-----	-----	-----	-----
-----	Leaf (interaction)	Light	2,314	1,011
			1,303	Root (interaction)

Light | 256 | 105 | 151 | | Root (additive fallback) | Light | 735 | 289 | 446 | | Combined
| Light | 460 | 302 | 158 |

Leaf shows a massive light effect with 2,314 significant DEGs -- 11.5% of all tested genes. This is expected: leaves are the primary light-sensing organ, and Blue vs Red light differentially regulates photosynthesis, photomorphogenesis, and secondary metabolism.

Root shows a much smaller light effect (256 DEGs from the interaction model; 735 from the better-powered additive model). Roots lack direct photoreception but receive light signals via systemic signaling from shoots.

Notable: In the combined model, only 460 genes are significant -- fewer than leaf alone. This reflects the dilution of tissue-specific light responses when pooling across tissues with fundamentally different light biology.

3.1.1 Leaf Light Effect: GO Enrichment

Blue-upregulated genes (604 Entrez IDs) were enriched for:

- Ethylene-activated signaling pathway (BP, padj=0.001)
- Regulation of flower development (BP, padj=0.001)
- Cellular response to phosphate starvation (BP, padj=0.002)

Blue-downregulated genes (841 Entrez IDs) were enriched for:

- Photosynthesis, light harvesting (BP, padj=5.2e-4)
- Proteolysis (BP, padj=7.7e-4)
- L-phenylalanine catabolic process (BP, padj=0.006)
- Regulation of photoperiodism, flowering (BP, padj=0.019)
- Structural constituent of chromatin (MF, padj=3.1e-5)
- Protochlorophyllide reductase activity (MF, padj=0.007)
- Nucleosome (CC, padj=8.5e-6)

The downregulation of light harvesting and protochlorophyllide reductase under Blue light is consistent with Blue light suppressing chlorophyll biosynthesis genes relative to Red light, which more strongly drives photosynthetic gene expression.

3.1.2 Root Light Effect: GO Enrichment

Blue-upregulated genes (173 Entrez IDs, additive model) were enriched for:

- Monooxygenase activity (MF, $\text{padj}=3.3\text{e-}4$)
- Oxidoreductase activity with oxygen incorporation (MF, $\text{padj}=3.7\text{e-}4$)
- Iron ion binding (MF, $\text{padj}=1.0\text{e-}3$)
- Heme binding (MF, $\text{padj}=7.6\text{e-}3$)
- Ethylene-activated signaling pathway (BP, $\text{padj}=0.097$)
- Recognition of pollen (BP, $\text{padj}=0.097$)

Blue-downregulated genes (265 Entrez IDs) were enriched for:

- Xyloglucan metabolic process (BP, $\text{padj}=0.024$)
- Cell wall biogenesis (BP, $\text{padj}=0.024$)
- Glucan metabolic process (BP, $\text{padj}=0.024$)
- Response to desiccation (BP, $\text{padj}=0.030$)
- Cell wall (CC, $\text{padj}=4.2\text{e-}3$)
- Apoplast (CC, $\text{padj}=1.3\text{e-}2$)

Root light responses center on cell wall remodeling and oxidoreductase activity, consistent with systemic light signals modulating root growth and cell wall properties.

3.1.3 KEGG Pathways: Light Main Effect

Leaf Blue>Red: 22 pathways enriched. Top pathways:

- Plant hormone signal transduction ($\text{padj}=8.2\text{e-}7$, 22 genes)
- Pentose phosphate pathway ($\text{padj}=1.1\text{e-}5$, 8 genes)
- Carbon fixation by Calvin cycle ($\text{padj}=1.5\text{e-}5$, 9 genes)

Leaf Blue<Red: 43 pathways enriched. Top pathways:

- Photosynthesis - antenna proteins ($\text{padj}=3.8\text{e-}7$, 9 genes)
- Plant-pathogen interaction ($\text{padj}=2.6\text{e-}6$, 20 genes)
- Porphyrin metabolism ($\text{padj}=2.6\text{e-}6$, 10 genes)

Root Blue>Red (additive): 8 pathways. Top: Plant hormone signal transduction (padj=4.2e-5), MAPK signaling (padj=1.5e-4).

Root Blue<Red: 10 pathways. Top: Arginine and proline metabolism (padj=4.5e-4), nucleotide sugar biosynthesis (padj=2.6e-3).

Shared leaf-root light pathways (11 total): Plant hormone signal transduction, MAPK signaling, Carotenoid biosynthesis, Flavonoid biosynthesis, Phosphatidylinositol signaling, Glycerophospholipid metabolism, Amino sugar metabolism, Galactose metabolism, Stilbenoid/diarylheptanoid biosynthesis, Glycerolipid metabolism, Biosynthesis of unsaturated fatty acids.

3.2 Light x Spaceflight Interaction

The interaction term identifies genes where the spaceflight response differs between Blue and Red light conditions. This is the central question of the analysis.

	Model	Effect	Strict (padj<0.05, LFC >=1)	Relaxed (padj<0.1, LFC >=0.5)	
Positive (Blue stronger)					
Negative (Red stronger)					
Leaf	Light x Condition	2,894	5,042	2,502	2,540
Root	Light x Condition	15	44	32	12
Combined	Light x Condition	845	1,900	725	1,175

Leaf shows a remarkably large interaction: 5,042 genes at relaxed thresholds (25% of all tested genes). The near-equal split between positive and negative interactions indicates that Blue light enhances the spaceflight response for roughly half of interaction genes while Red light enhances it for the other half.

Root shows very few interaction genes (44 at relaxed thresholds), consistent with the low power from only 3 ground replicates per light condition. These results should be treated as exploratory.

Combined shows 1,900 interaction genes, with more negative interactions (1,175) than positive (725), suggesting that across the whole plant, Red light tends to amplify the spaceflight response more than Blue light.

3.2.1 Leaf Interaction: GO Enrichment

Positive interaction (flight effect stronger under Blue, 1,675 Entrez IDs):

- Proteolysis (BP, $\text{padj}=0.026$)
- Aromatic amino acid family biosynthetic process (BP, $\text{padj}=0.084$)

Negative interaction (flight effect stronger under Red, 1,573 Entrez IDs):

- Response to heat (BP, $\text{padj}=2.2\text{e-}3$)
- Ethylene-activated signaling pathway (BP, $\text{padj}=3.2\text{e-}3$)
- Xyloglucan metabolic process (BP, $\text{padj}=0.068$)
- Regulation of flower development (BP, $\text{padj}=0.068$)
- Cellular response to phosphate starvation (BP, $\text{padj}=0.068$)
- Heat shock protein binding (MF, $\text{padj}=0.015$)
- Protein-folding chaperone binding (MF, $\text{padj}=0.034$)
- Cell wall (CC, $\text{padj}=0.045$)

The negative interaction enrichment for ethylene signaling and heat shock proteins is particularly notable: these stress-response pathways show stronger spaceflight induction under Red light than Blue light.

3.2.2 Leaf Interaction: KEGG Enrichment

Positive interaction (83 pathways): Top pathways include Biosynthesis of cofactors ($\text{padj}=3.1\text{e-}10$), Biosynthesis of amino acids ($\text{padj}=6.9\text{e-}10$), Plant-pathogen interaction ($\text{padj}=7.8\text{e-}9$).

Negative interaction (48 pathways): Top pathways include:

- **Plant hormone signal transduction** ($\text{padj}=1.0\text{e-}13$, 52 genes)
- **Protein processing in ER** ($\text{padj}=5.4\text{e-}10$, 31 genes)
- **MAPK signaling pathway - plant** ($\text{padj}=5.5\text{e-}8$, 26 genes)

The negative interaction for hormone signaling and MAPK signaling means that the spaceflight-induced upregulation of these pathways (identified in the previous additive analysis) is **stronger under Red light than Blue light**. This is a key finding: Red light amplifies the spaceflight stress response in hormone and MAPK signaling.

3.2.3 Root Interaction: Enrichment (Exploratory)

Root interaction gene sets are very small (9-16 Entrez IDs), so enrichment results are driven by single genes and should be interpreted cautiously.

Positive interaction (16 Entrez IDs): Response to photooxidative stress (BP), Photosystem I assembly (BP), Channel activity (MF), Clathrin-coated vesicle (CC).

Negative interaction (9 Entrez IDs): Ethylene-activated signaling pathway (BP), Hydrogen peroxide catabolic process (BP), Response to oxidative stress (BP), Peroxidase activity (MF).

Despite the small gene sets, the recurrence of ethylene signaling and oxidative stress in root interaction results mirrors the leaf findings, suggesting these may be genuine cross-tissue interaction effects.

3.2.4 Combined Interaction: KEGG Enrichment

Positive interaction (498 Entrez IDs, 52 pathways): Top pathways:

- **Phenylpropanoid biosynthesis** ($p_{\text{adj}}=4.2\text{e-}10$, 19 genes)
- Phenylalanine metabolism ($p_{\text{adj}}=3.8\text{e-}7$, 10 genes)
- Biosynthesis of amino acids ($p_{\text{adj}}=4.3\text{e-}7$, 17 genes)

Negative interaction (715 Entrez IDs, 23 pathways): Top pathways:

- Protein processing in ER ($p_{\text{adj}}=1.7\text{e-}4$, 14 genes)
- Plant hormone signal transduction ($p_{\text{adj}}=1.7\text{e-}4$, 20 genes)
- Phosphatidylinositol signaling system ($p_{\text{adj}}=2.8\text{e-}4$, 8 genes)

The combined model confirms the leaf pattern: phenylpropanoid biosynthesis shows a positive interaction (spaceflight effect stronger under Blue), while hormone signaling shows a negative interaction (spaceflight effect stronger under Red).

3.3 Systems Biology Pathway Visualization

3.3.1 Pathway LFC Heatmap

A comprehensive heatmap of mean LFC across 10 key KEGG pathways x 6 comparisons reveals striking tissue-specific patterns:

Pathway	Leaf: Blue vs Red	Leaf: Light x Flight	Root: Blue vs Red	Root: Light x Flight
Photosynthesis - antenna proteins	-1.97	+1.54	+0.07	+0.00
Porphyrin metabolism	-0.66	+0.68	+0.01	+0.00
Phenylpropanoid biosynthesis	-0.31	+0.65	+0.22	+0.04
Carbon fixation (Calvin cycle)	+0.33	-0.24	+0.01	+0.00
Plant hormone signal transduction	+0.02	-0.13	+0.02	+0.00
MAPK signaling - plant	-0.02	-0.06	+0.04	+0.00
Circadian rhythm - plant	-0.05	-0.01	+0.01	+0.00
Flavonoid biosynthesis	+0.03	+0.49	+0.01	+0.00
Photosynthesis	-0.27	+0.29	+0.02	+0.00
Protein processing in ER	+0.01	-0.23	-0.01	+0.00

Key observations:

1. **Photosynthesis - antenna proteins** shows the strongest light effect (-1.97 under Blue vs Red) AND the strongest interaction (+1.54). This means antenna protein genes are strongly downregulated by Blue light relative to Red, but the spaceflight response (which upregulates these genes) is much stronger under Blue light -- partially compensating for the Blue-light downregulation.
2. **Porphyrin metabolism** (chlorophyll biosynthesis precursors) mirrors the antenna protein pattern exactly.
3. **Phenylpropanoid biosynthesis** shows a positive interaction (+0.65) -- the spaceflight-induced upregulation of phenylpropanoid genes is enhanced under Blue light.
4. **Root pathways show near-zero interaction effects**, consistent with the low power and potentially genuine absence of light x spaceflight interaction in root

tissue.

3.3.2 SBGN-style Pathway Diagrams

SBGN-style Process Description diagrams were generated for three key pathways using KEGG KGML spatial layouts with SBGN visual conventions:

- **Plant hormone signal transduction** (sly04075): Shows gene-level LFC coloring for both light effect and interaction
- **Phenylpropanoid biosynthesis** (sly00940): Highlights the positive interaction in secondary metabolism
- **Circadian rhythm - plant** (sly04712): Shows modest but consistent light effects on clock genes

These diagrams use rounded rectangles (SBGN macromolecule convention) for genes/proteins and circles (simple chemical convention) for metabolites, with color intensity representing LFC magnitude.

3.4 Tissue-Specific vs Shared Effects

3.4.1 Gene-Level Overlap

Comparison	Leaf genes	Root genes	Shared	% of union	Direction reversals
Light up (Blue>Red)	1,011	289	37	2.9%	38 (leaf up & root down)
Light down (Blue<Red)	1,303	446	28	1.6%	40 (leaf down & root up)
Interaction positive	2,502	32	1	~0%	--
Interaction negative	2,540	12	2	0.1%	--

The overlap between leaf and root light-effect genes is very small (2-3% of the union). Remarkably, the number of **direction reversals** (38-40 genes where Blue light upregulates in leaf but downregulates in root, or vice versa) exceeds the number of

consistently regulated shared genes (65). This indicates fundamentally different light responses in the two tissues.

3.4.2 LFC Correlation Between Tissues

Comparison	Pearson r	Spearman rho	n genes
Light effect (Blue vs Red)	-0.032	-0.063	19,529
Light x Flight interaction	0.008	0.042	19,529

The near-zero correlations confirm that leaf and root have essentially independent transcriptional responses to light quality and to the light x spaceflight interaction. This is biologically expected: leaves directly perceive light via photoreceptors, while roots receive only indirect systemic signals.

3.4.3 Within-Tissue: Light Effect vs Interaction Correlation

Tissue	Pearson r (Light LFC vs Interaction LFC)
Leaf	-0.840
Root	-0.282

The leaf shows a striking **negative correlation (r = -0.840)** between the light main effect and the light x flight interaction. This means: genes that are strongly upregulated by Blue light (relative to Red) tend to show a weaker spaceflight response under Blue light, and vice versa. This is consistent with a **compensatory model**: Blue light pre-activates certain pathways (e.g., photosynthesis, chlorophyll biosynthesis), reducing the additional perturbation caused by spaceflight on those same pathways. Conversely, pathways suppressed by Blue light (e.g., antenna proteins) show stronger spaceflight induction under Blue -- as if the spaceflight environment partially rescues the Blue-light suppression.

The root shows a weaker but still negative correlation (r = -0.282), suggesting a muted version of the same compensatory dynamic.

3.4.4 Shared vs Tissue-Specific KEGG Pathways

Light effect: 11 KEGG pathways are shared between leaf and root (out of 65 total). Shared pathways include Plant hormone signal transduction, MAPK signaling, Carotenoid biosynthesis, and Flavonoid biosynthesis -- all pathways with known systemic signaling components.

Interaction: Only 4 pathways are shared (all driven by single root genes): Phenylpropanoid biosynthesis, Lysosome biogenesis, Carotenoid biosynthesis, and Riboflavin metabolism. The leaf dominates interaction enrichment with 104 leaf-only pathways.

3.5 Comparison with Previous Flight-vs-Ground Results

The previous additive model ($\sim \text{light} + \text{condition}$) identified 2,132 leaf DEGs and 2,582 root DEGs for the Flight vs Ground comparison. The interaction model reveals that:

1. **The condition main effect changes substantially** when the interaction is included: leaf condition DEGs drop from 2,132 (additive) to 527 (interaction model, strict threshold). This is because the additive model conflates the main spaceflight effect with the interaction -- some genes that appeared as "spaceflight-responsive" were actually responding differently depending on light color.
 2. **The light main effect is much larger than the spaceflight effect in leaf:** 2,314 light DEGs vs 527 condition DEGs. Light quality is the dominant transcriptional regulator in photosynthetic tissue.
 3. **The interaction is comparable in magnitude to the main effects:** 5,042 interaction genes (relaxed) vs 2,314 light and 527 condition. This means light quality substantially modulates how plants transcriptionally respond to spaceflight.
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4. Discussion

4.1 Light Quality as a Modulator of Spaceflight Response

The central finding is that **light quality (Red vs Blue) profoundly modulates the spaceflight transcriptomic response in tomato leaf**, affecting approximately 25%

of all expressed genes. The interaction is largely leaf-specific, with minimal detectable interaction in root (though this may reflect limited power rather than genuine absence).

4.2 The Compensatory Model

The strong negative correlation ($r = -0.840$) between light main effect and interaction LFC in leaf supports a compensatory model: pathways that are strongly activated by one light condition show attenuated spaceflight responses under that same condition. This could reflect:

1. **Saturation effect:** If Blue light already maximally activates photosynthesis-related genes, spaceflight cannot further upregulate them.
2. **Pre-adaptation:** Light-condition-specific gene expression may pre-adapt plants to certain stresses, reducing the additional perturbation from microgravity.
3. **Resource allocation trade-offs:** Plants under Blue light may allocate resources differently, leaving fewer reserves for spaceflight-induced changes in Blue-activated pathways.

4.3 Antenna Proteins: A Case Study in Light x Spaceflight Interaction

The Photosynthesis - antenna proteins pathway provides the clearest example:

- **Blue vs Red:** mean LFC = -1.97 (strong downregulation under Blue)
- **Light x Flight interaction:** mean LFC = +1.54 (spaceflight upregulation is much stronger under Blue)

Under Red light, antenna protein genes are highly expressed and show minimal spaceflight response. Under Blue light, these genes are suppressed but show strong spaceflight-induced upregulation -- as if microgravity partially rescues the Blue-light suppression of light-harvesting machinery. This may represent an adaptive response: in microgravity, where photosynthetic demand may change, plants under Blue light upregulate antenna proteins to compensate for reduced light harvesting efficiency.

4.4 Hormone Signaling: Red Light Amplifies Spaceflight Stress

Plant hormone signal transduction shows a **negative interaction** (stronger spaceflight response under Red light) in both leaf and combined models. This suggests that Red light amplifies the spaceflight-induced disruption of hormone signaling (auxin, ethylene, cytokinin pathways). Given that phytochromes (Red light receptors) directly interact with hormone signaling cascades (particularly auxin transport), this interaction may reflect a phytochrome-hormone crosstalk that is disrupted in microgravity.

4.5 Phenylpropanoid Biosynthesis: Blue Light Enhances Spaceflight-Induced Secondary Metabolism

Phenylpropanoid biosynthesis shows a **positive interaction** (stronger spaceflight response under Blue light). This pathway produces flavonoids, lignins, and other secondary metabolites involved in UV protection and stress responses. Blue light (which includes wavelengths closer to UV) may prime the phenylpropanoid pathway, making it more responsive to the additional oxidative stress of spaceflight.

4.6 Tissue Specificity

The near-zero correlation between leaf and root LFCs for both light effects and interactions underscores that these are fundamentally different tissue environments:

- **Leaf** directly perceives light and shows massive light and interaction effects
- **Root** receives only indirect light signals and shows minimal interaction effects
- The few shared pathways (hormone signaling, MAPK) likely reflect systemic signaling rather than direct light perception

4.7 Limitations

1. **Root power:** Only 3 ground replicates per light condition severely limits interaction detection in root. The 44 root interaction genes (relaxed threshold) should be considered exploratory.
2. **Unbalanced design:** Flight Blue has only 4 samples per tissue, which may inflate interaction estimates.

3. **Gene ID mapping:** Only ~60% of Solyc IDs map to Entrez IDs, reducing enrichment power and potentially introducing mapping bias.
 4. **pathview/SBGNview unavailable:** The Rgraphviz dependency could not be compiled due to graphviz API version incompatibility. Alternative visualizations were generated but lack the polished pathway overlay of pathview.
 5. **Interaction threshold:** The relaxed threshold ($p_{adj} < 0.1$, $|LFC| \geq 0.5$) for interaction genes may include some false positives. The large interaction count (5,042 in leaf) should be interpreted with this in mind.
 6. **No replication across experiments:** These results are from a single spaceflight experiment. Validation in independent experiments would strengthen the conclusions.
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5. Conclusions

1. **Light quality profoundly modulates the spaceflight transcriptomic response** in tomato leaf, with ~25% of genes showing significant light x spaceflight interaction.
2. **A compensatory dynamic exists:** genes strongly regulated by light quality tend to show attenuated spaceflight responses under the same light condition ($r = -0.840$ in leaf).
3. **Antenna proteins and porphyrin metabolism** show the strongest interaction: Blue light suppresses these pathways, but spaceflight strongly upregulates them under Blue light.
4. **Hormone signaling and MAPK pathways** show stronger spaceflight induction under Red light, suggesting Red light amplifies spaceflight stress signaling.
5. **Phenylpropanoid biosynthesis** shows stronger spaceflight induction under Blue light, consistent with Blue light priming secondary metabolism for stress responses.
6. **Leaf and root have essentially independent light x spaceflight responses** ($r \sim 0$), reflecting their different roles in light perception and systemic signaling.
7. **These findings have practical implications:** light environment design in spaceflight growth chambers could be optimized to mitigate specific aspects of spaceflight stress -- e.g., Blue light supplementation may reduce excessive

hormone signaling disruption, while Red light may help maintain photosynthetic capacity.

6. Output Files

DESeq2 Results

- `leaf_light_Blue_vs_Red.csv` -- Leaf light main effect (20,157 genes)
- `leaf_interaction.csv` -- Leaf light x condition interaction
- `root_light_Blue_vs_Red.csv` -- Root light effect (interaction model)
- `root_light_additive_Blue_vs_Red.csv` -- Root light effect (additive fallback)
- `root_interaction.csv` -- Root light x condition interaction
- `combined_light_Blue_vs_Red.csv` -- Combined light main effect
- `combined_interaction.csv` -- Combined light x condition interaction
- `deseq2_summary_table.csv` -- Summary of all DESeq2 results

Enrichment Results

- `go_enrichment_all.csv` -- All GO enrichment results (137 terms)
- `kegg_enrichment_all.csv` -- All KEGG enrichment results (313 pathways)
- `gene_set_summary.csv` -- Gene set sizes and mapping rates

Tissue Comparison

- `tissue_overlap_genes.csv` -- Leaf vs root gene set overlap
- `tissue_lfc_correlation.csv` -- Per-gene LFCs for correlation analysis
- `shared_light_up_genes.csv` -- Genes upregulated by Blue in both tissues
- `shared_light_down_genes.csv` -- Genes downregulated by Blue in both tissues

Figures

- `leaf_light_MA.png`, `leaf_interaction_MA.png` -- MA plots
- `leaf_light_volcano.png`, `leaf_interaction_volcano.png` -- Volcano plots
- `root_light_MA.png`, `root_interaction_MA.png` -- Root MA plots

- `root_light_volcano.png`, `root_interaction_volcano.png` -- Root volcano plots
- `combined_light_MA.png`, `combined_interaction_MA.png` -- Combined MA plots
- `combined_light_volcano.png`, `combined_interaction_volcano.png` -- Combined volcano plots
- `leaf_light_vs_interaction_lfc.png` -- Within-leaf LFC correlation ($r=-0.840$)
- `root_light_vs_interaction_lfc.png` -- Within-root LFC correlation ($r=-0.282$)
- `tissue_comparison_venn_scatter.png` -- Venn diagrams + LFC scatter plots
- `kegg_tissue_comparison_heatmap.png` -- KEGG pathway enrichment heatmap
- GO dotplots: `leaf_light_up_BP_dotplot.png`, `leaf_light_down_BP_dotplot.png`, etc.

Pathway Visualization (in `pathways/` subdirectory)

- `pathway_lfc_heatmap.svg/.png` -- Pathway-level LFC heatmap (10 pathways x 6 comparisons)
- `pathway_expression_summary.csv` -- Pathway expression data
- `sly04075_gene_lfc_bars.svg/.png` -- Plant hormone signal transduction gene-level plot
- `sly00940_gene_lfc_bars.svg/.png` -- Phenylpropanoid biosynthesis gene-level plot
- `sly00196_gene_lfc_bars.svg/.png` -- Photosynthesis antenna proteins gene-level plot
- `sly04712_gene_lfc_bars.svg/.png` -- Circadian rhythm gene-level plot
- `sly04075_composite.svg/.png` -- Hormone signaling: KEGG image + expression table
- `sly04075_sbgn_style.svg/.png` -- SBGN-style hormone signaling diagram
- `sly00940_sbgn_style.svg/.png` -- SBGN-style phenylpropanoid diagram
- `sly04712_sbgn_style.svg/.png` -- SBGN-style circadian rhythm diagram
- KEGG reference images for all 10 pathways